Amrita Kadam

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Education

Indian Institute of Information Technology, Raichur **Expected Aug 2027**
B.Tech in Computer Science and Engineering *Raichur, India*

- **Mathematics:** Probability and Statistics, Linear Algebra, Calculus (Engineering Mathematics I–III), Discrete Mathematics.
- **Machine Learning:** Foundations of Machine Learning, Deep Learning, Natural Language Processing, Agentic AI. **Computing:** Data Structures and Algorithms.

Experience

Moleculyst **Jul 2026 – Present**
Mechanistic Interpretability Research Intern | *github.com/Amrita0205/multihaludet-reproduction* *Remote*

- Reproduced **MultiHaluDet** (ACL 2026 workshop), which detects **LLM hallucinations** by probing a frozen model's hidden states: matched its HaluEval result on Mistral-7B (**AUROC 0.985** vs. 0.984 reported), then extended it to **Qwen2.5-7B, Phi-3.5-mini and Yi-1.5-9B**. All four scored 0.985–0.989; the 3.8B Phi-3.5 matched the 7–9B models at half the GPU memory.
- **Found a validation bug** in the released code: a cross-validation seed collision **inverted the meta-learner's predictions** (AUROC 0.015); decorrelating the folds restored out-of-fold AUROC from 0.41 to 0.98. Added failure accounting and feature health checks so an incompatible model fails loudly instead of producing plausible-looking scores.
- Wrote a **GPU meta-learner** that trains in **51 s** vs. ∼**50 min** for the original ensemble, so many LLMs can be compared.

TerraBiz **Feb 2025 – Aug 2025**
Backend Developer Intern *Remote*

- Built and deployed **production REST APIs** in Node.js and Express.js.
- Reduced query latency by restructuring **MongoDB** schemas and adding targeted indexes.

Projects

Veda AI — Assessment Extraction and Answer Mapping **2026**
github.com/Amrita0205/Extraction-and-Answer-Mapping-Veda-AI | Live app

- Shipped a **deployed vision-LLM pipeline** (**FastAPI**, **Gemini** vision, Next.js) that reads a printed question paper and a handwritten answer sheet, then extracts the questions, maps each answer to its question, highlights its exact ink region and grades it.
- Designed a **five-stage mapping ladder** (written label, embedded label, IDF-weighted overlap, model adjudication, sequence) so assignment is **order-independent**: out-of-sequence answers, split sub-parts and page-spanning answers all resolve.
- **Debugged a silent failure**: a strict length check rejected valid bounding boxes and fell back to plausible-looking whole-page highlights instead of erroring. Snapping boxes to ink took overlap from **IoU 0.29 to 0.89** and whole-page fallbacks to **0 of 52 regions**; cut page extraction latency from **176s to 2.2s**.
- Built a **synthetic ground-truth harness** (pytest) and a **14-subject evaluation sweep**: **F1 1.00** extraction on the synthetic sheet, **90% mean extraction coverage** across subjects, and **16 of 17** answers correctly mapped on an independently labelled sheet whose **35/40 grade the pipeline matched exactly**.

Agentic AI Career Advisor **Nov 2025**
github.com/anandn1/career-advisor | youtu.be/56Xtd1PNetw

- Team of four. Wrote the **supervisor** that orchestrates the **LangGraph** agents (job-market analysis, skill-gap detection, resume strategy, mock interviews, learning-path planning) over a shared session state, taking the system to a working end-to-end pipeline.
- Advice is grounded with **RAG** over **ChromaDB** (embeddinggemma-300m); selected by **IIIT Dharwad faculty** for research collaboration.

Blog Writing Agent — Multi-Agent Content Pipeline **2026**
github.com/Amrita0205/Blog_writing_AI_agent

- Built a four-agent **CrewAI** pipeline (Researcher, Writer, Editor, Social) producing a publication-ready post plus platform variants on **Groq llama-3.3-70b**; resolved a `litellm/openai` dependency conflict and added rate-limit retries for stable runs.

Achievements and Leadership

- Selected for the **Summer of Bitcoin** 2026 developer challenge round, a global open-source program; built a Bitcoin transaction analyser and a BIP-174 PSBT builder.
- **Tech Coordinator, CodeSoc, IIIT Raichur** (2024–2025): mentored 150+ students in data structures and algorithms; organised the Code Conflux hackathon and the InterIIIT UDBHAV round.

Technical Skills

Languages: Python, SQL, C++
ML and Deep Learning: PyTorch, **Hugging Face Transformers**, scikit-learn, NumPy, pandas
Evaluation and Probing: hallucination detection, **hidden-state probing**, cross-validation, AUROC / F1 / IoU evaluation, pytest
LLMs and Agents: LangGraph, LangChain, CrewAI, **RAG**, ChromaDB, vector search, Gemini vision, Groq
Engineering and Tools: FastAPI, Django, REST APIs, PostgreSQL, MongoDB, Git, GitHub Actions (CI/CD)